

KHOI BUI DINH

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EDUCATION

2020-2024
(completed) **B.S. Computer Science**
Minor in Integrated Sciences
Fulbright University Vietnam (Ho Chi Minh City, Vietnam)
cGPA: 3.70/4.00

EXPERIENCES

Jan 2025-
Present **Bioinformatics Technician**, Pacifics Informatics (Ho Chi Minh City, Vietnam)

- o Developing & maintaining Nextflow pipelines for human, viral & chimeric reads extraction from HPV-infected cancer patients, benchmarking bioinformatics tools for the pipeline integration.
- o Adapting aladdin-viralrecon-illumina: a nf-core-based pipeline for Zymo's aladdin platform to perform assembly and intra-host/low-frequency variant calling for viral samples.
- o Constructing biofetch - Nextflow pipeline to parallelly retrieve FASTQ and metadata from NCBI, ENA.

June 2024-
Nov 2024 **Collaborator**, BioDev Network's Future Innovators Mentorship Program, Wellcome Trust Sanger (Remote)

Project: Responsible AI for Genomics Research

Advisor: Dr. Priyanka Surana

- o Revising literature on AI models and their applications in cancer drug development stages.
- o Designing and implementing a benchmark criteria to evaluate models' FAIRness and robustness.
- o Improving models' explainability by applying SHAP and LIME tools on existing models, incorporating Nextflow to enhance models' development pipeline.

June 2024-
Aug 2024 **Bioinformatics Intern**, BioTuring (Ho Chi Minh City, Vietnam)

- o Pre-processed single-cell data utilizing Python and Bash Scripting, performed cell type annotation using heatmap and marker genes in scRNA, perturb-seq and spatial transcriptomics experiments.
- o Created notebooks pipeline for merging samples for scRNA, spatial transcriptomics, and Perturb-seq experiments.

Feb 2024-
May 2024 **Undergraduate Research Assistant**, Fulbright University Vietnam (Ho Chi Minh City, Vietnam)

Project: Detection and classification of coral-reef species in Dam Tre, Vietnam

Advisor: Dr. Jesse Hollister and Dr. Phan Thanh Trung

- o Extracted, annotated and pre-processed coral reef images gathered from Dam Tre, Vietnam.
- o Performed web scraping to gather coral reef datasets from CoralNet for model training.
- o Revised literature on Deep Learning techniques for object detection and classification.

Jan 2023-
Jan 2024 **Undergraduate Research Assistant**, Fulbright University Vietnam (Ho Chi Minh City, Vietnam)

Project: Influences of fanatical beliefs in the co-evolution of networks and beliefs

Advisor: Dr. Graeme Walker

- o Developed a computational model to simulate and visualize the co-evolution of networks & beliefs using Python and NetworkX.
- o Analyzed the effects of introducing fanatical beliefs into a population of 200 people, captured the results when the network diverged and evolved over a 100+ period span.

Sep 2023-
Dec 2023 **Teaching Assistant**, Fulbright University Vietnam (Ho Chi Minh City, Vietnam)

Course: Computer Science I

- o Assisted 30+ students by hosting weekly office hours for drop-ins, hosting revision workshops.
- o Graded students' homeworks on a weekly basis, and provided feedback for students' exams and projects.

- Mar 2023-
Sep 2023
- Full-stack Developer Intern**, Cosmodern Co. (Ho Chi Minh City, Vietnam)
- o Developed and restructured the website interface using Figma, JavaScript, SCSS, and HTML, leading to 10+ brand partners' satisfaction on revisit.
 - o Created and improved existing features, including coupon usage, order purchase, user registration, and authentication.

PRESENTATION

- Nov 2024 **Genome Informatics Conference**, Poster Presentation (Wellcome Genome Campus, Cambridge)
Bui, K., Loay, O., Garnham, M. D., & Surana, “Responsible AI for Genomics Data Analysis”
- Oct 2024 **Explainable AI in Biology Conference**, Poster Presentation (Wellcome Genome Campus, Cambridge)
Bui, K., Loay, O., Garnham, M. D., & Surana, “Responsible AI in Genomics Data Analysis”

RELEVANT SKILLS

Languages	Python, R, Java, JavaScript, PHP, SQL, HTML, CSS
Frameworks	Quarto, Laravel, React-Bootstrap, Shiny, PyTorch, Tensorflow, nf-core
Tools	Nextflow, Docker, Jupyter, Git, Firebase, Bash, MongoDB, CellRanger, HPC
Softwares	Microsoft Office (Word, Excel, Presentation), Canva, Figma, AzureData Studio, Latex
Soft skill	Scientific Writing, Scientific Communication, Research, Project Management, Public Presentation

AWARDS & ACHIEVEMENTS

- Mar 2025 **Contributor**, Nextflow Hackathon 2025, [scxa-tertiary-workflow](#)
- Dec 2024 **8.0**, IELTS
- Mar 2023 **Scholarship Recipient**, Grow with Google - Accelerate Vietnam Digital Talent by NIC x Google
- Nov 2022 **Top 30**, RMIT Business Analyst Champions

LEADERSHIP & ACTIVITIES

- May 2021-
Aug 2021
Part-time
- Co-designer**, Mindful-bright - Fulbright’s mindfulness initiative (Ho Chi Minh City, Vietnam)
- o Participated in a training workshop for co-designing an initiative.
 - o Started and co-designed Mindful-bright with the Presidential Office to propose mindfulness practices to students of the Fulbright community.
- Jan 2020-
Jan 2021
Part-time
- Local Branch Vice President**, AIESEC in Vietnam (Ho Chi Minh City, Vietnam)
- o Led the AIESEC local branch for human resources & financial management, provided leadership training and managed recruitment projects.
- Feb 2020-
July 2020
Part-time
- National Supporting Team Member**, AIESEC in Vietnam (Ho Chi Minh City, Vietnam)
- o Collaborated & delivered national financial tracking & forecasting tools.
 - o Performed national training on financial management & sustainability for local projects.
- Jun 2019-
Aug 2019
Part-time
- Exchange Participant**, AIESEC in India (Jodhpur City, India)
- o Implemented English teaching, food sustainability & cultural exchange projects in Jodhpur and Ahmedabad.
 - o Collaborated & delivered “World’s Largest Lesson” to schools regarding SDG4 - Quality Education.

REFERENCES

Priyanka Surana, Ph.D.	Senior Scientific Manager , Scientific Solutions Wellcome Trust Sanger ps22@sanger.ac.uk
Graeme Walker, Ph.D.	Associate Dean , Faculty Affairs Fulbright University Vietnam graeme.walker@fulbright.edu.vn
Linh Huynh, Ph.D.	Faculty , Computer Science Fulbright University Vietnam linh.huynh@fulbright.edu.vn